Highlights

WaveSplat: Grid-Free Spectral Wave Field Estimation from Sparse Shipboard LiDAR

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- Grid-free spectral wave estimation from sparse shipboard LiDAR
- 51.5% lower nRMSE than ETKF-grid, 2.4× faster (13.4 ms/frame)
- Dispersion-constrained parameterization ensures physical consistency
- Adaptive model order matches spectral complexity to observed scene
- Robust to both diffuse and specular LiDAR observation models

WaveSplat: Grid-Free Spectral Wave Field Estimation from Sparse Shipboard LiDAR

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Abstract

We present WaveSplat, a grid-free method for real-time ocean wave field estimation from sparse shipboard LiDAR observations. Unlike grid-based approaches such as the Ensemble Transform Kalman Filter (ETKF) that discretize the spatial domain and require interpolation of sparse observations onto regular grids, WaveSplat represents the wave field as a superposition of spectral components—plane waves parameterized by wavenumber, amplitude, and phase—with angular frequency constrained by the linear dispersion relation. This continuous spectral representation avoids interpolation artifacts, scales to arbitrary spatial resolution, and enables adaptive model complexity matched to the true wave scene.

We propose a two-phase estimation pipeline: (1) initialization via dispersion-aware beamforming with coarse-to-fine refinement, and (2) per-frame tracking with physics-based phase prediction and selective gradient updates. A Fresnel-zone confidence weighting scheme accounts for the range- and angle-dependent noise characteristics of LiDAR measurements on ocean surfaces.

On 12 representative episodes spanning Beaufort 2–5 sea states (1–30 wave components), WaveSplat achieves an average normalized RMSE (nRMSE) of **0.211** compared to 0.436 for ETKF-grid (**51.5%** improvement), while running **2.4**× faster (13.4 vs. 31.7 ms/frame). A multi-seed statistical validation (50 paired runs) confirms significance ($p < 0.001$, Wilcoxon signed-rank test). The

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method is robust to both diffuse and specular LiDAR observation models, winning on all 20 paired test episodes. With adaptive model order, the method matches spectral complexity to the observed scene—selecting N=2 for simple two-component swells and N=18–21 for complex multi-component seas—achieving 6.4% average improvement over fixed model order. (At very low sea states where signal-to-noise is poor, the adaptive mechanism can overfit; this failure mode is analyzed in Section 6.)

Keywords: wave field estimation, LiDAR, spectral methods, dispersion relation, data assimilation, real-time estimation

Table 1: Nomenclature.

Symbol	Description
$z(x, y, t)$	Surface elevation [m]
N	Number of spectral components
w_i	Amplitude of i-th component [m]
$k_{x,i}, k_{y,i}$	Wavenumber components [rad/m]
ω_i	Angular frequency [rad/s]
φ_i	Phase offset [rad]
g	Gravitational acceleration [m/s^2]
d	Water depth [m]
H_s	Significant wave height [m]
T_p	Peak period [s]
c_j	Confidence weight for observation j
θ_j	Beam incidence angle [rad]
M	Number of LiDAR observations per frame
nRMSE	RMSE normalized by H_s

1. Introduction

1.1. Motivation: wave field information for maritime autonomy

Knowledge of the ocean wave field surrounding a vessel is a prerequisite for many functions in autonomous and semi-autonomous ship operations [1, 2]. Short-term wave prediction enables proactive motion compensation for cargo handling and helicopter landing on offshore platforms [3]. Route optimization algorithms require spatially resolved wave information to minimize fuel consumption and structural fatigue loading [4]. Active ride control systems—particularly relevant for high-speed hydrofoil vessels—can exploit wave preview for feedforward disturbance rejection [5]. Structural health monitoring benefits from accurate wave load estimation for fatigue life assessment [6]. In each application, the value of wave field information depends on its spatial resolution, temporal update rate, and accuracy.

1.2. Shipboard LiDAR for ocean surface sensing

Several sensor modalities have been employed for sea surface observation. Wave buoys provide accurate point measurements but lack spatial resolution [7]. X-band marine radar can image the sea surface over large areas ($\sim$2–4 km radius) but measures backscatter intensity rather than elevation directly, requiring empirical or physics-based inversion to recover wave spectra [8, 9]. Satellite altimetry provides global coverage at coarse spatial resolution, unsuitable for real-time vessel-scale applications [10].

Shipboard scanning LiDAR has emerged as a compelling alternative, providing *direct* elevation measurements over a wide area at high frame rates ($\sim$10 Hz) [11, 12]. Modern solid-state LiDAR systems (e.g., Seyond Falcon K1: 1550 nm, 152 channels, $120^\circ \times 25^\circ$ FOV) can measure ocean surface topography at ranges up to 250 m with centimeter-level accuracy in the near field. However, the observations are inherently sparse (typically 50–200 valid surface returns per scan), unevenly distributed (denser near the ship due to the fixed angular resolution), and noisy (accuracy degrades with range due to beam divergence, and with incidence angle due to specular reflection on the water surface).

The challenge is therefore to reconstruct a continuous, physically consistent wave field from these sparse, heterogeneous observations in real time.

1.3. Grid-based paradigm and its limitations

The dominant approach to this reconstruction problem follows a grid-based data assimilation paradigm, adapted from numerical weather prediction: discretize the spatial domain into a regular grid, represent the wave state as surface elevation and velocity at grid points, and apply an ensemble Kalman filter (EnKF/ETKF) to sequentially assimilate observations [13, 14, 15]. The ensemble forecast step propagates the wave state forward in time using FFT-based spectral methods, which efficiently implement the linear dispersion relation on regular grids.

While principled and well-studied, this grid-based approach has three fundamental limitations when applied to sparse LiDAR observations:

1. **Grid interpolation artifacts**: Sparse, irregularly distributed LiDAR observations must be mapped to grid points via interpolation or observation operators, introducing smoothing for wavelengths near the grid Nyquist limit and artificial correlations between neighboring grid cells. For a typical 40×30 grid covering 200×150 m, the grid spacing is 5 m, limiting the minimum resolvable wavelength to ∼10 m regardless of the LiDAR point density in the near field.
2. **Fixed resolution**: The grid resolution is a design parameter that trades accuracy against computational cost. A fine grid (80×60) improves resolution but quadruples the ensemble state dimension. Crucially, the grid resolution is fixed regardless of whether the actual wave field contains one or thirty spectral components, wasting computational resources on simple sea states and under-resolving complex ones.
3. **Computational overhead**: The ensemble forecast requires FFT-based spectral propagation on the full grid for each of N_e ensemble members, with cost scaling as $O(N_e \cdot G^2 \log G^2)$ regardless of the true wave com-

plexity. With typical values ($N_e = 50$, $G = 40 \times 30$), each forecast step involves 50 independent 2D FFTs.

1.4. Proposed approach

We propose **WaveSplat**, a grid-free spectral representation that eliminates the spatial grid entirely. Instead of discretizing space, we represent the wave field as a weighted superposition of N plane waves:

$$z(x, y, t) = \sum_{i=1}^{N} w_i \cos(k_{x,i}x + k_{y,i}y - \omega_i t + \varphi_i) \tag{1}$$

where each spectral component is parameterized by amplitude w_i, wavenumber $(k_{x,i}, k_{y,i})$, and phase φ_i. The angular frequency ω_i is not a free parameter but is *constrained* by the linear dispersion relation:

$$\omega_i = \sqrt{g\,|k_i|\,\tanh(|k_i|\,d)} \tag{2}$$

where $g = 9.81$ m/s^2 and d is the water depth. This hard constraint reduces the parameter space from $5N$ to $4N$, regularizes optimization, and ensures physically plausible wave propagation during temporal tracking.

Our key contributions are:

1. **Grid-free spectral estimation framework (C1)**: Direct estimation on sparse, irregularly distributed observations without grid interpolation, with frame-to-frame temporal tracking via dispersion-based phase propagation. Unlike single-snapshot sparse spectral methods [16], WaveSplat maintains component identity across frames, enabling continuous spatial evaluation at arbitrary resolution while naturally adapting to any observation pattern.
2. **Dispersion-constrained parameterization (C2)**: Hard enforcement of the full dispersion relation (including tanh depth dependence) reduces the free parameter count and acts as a physics-based regularizer, improving tracking stability over extended time horizons.

Table 2: Comparison of ocean surface sensing modalities for vessel-scale wave field estimation. Typical values for operational systems.

Sensor	Spatial range	Resolution	Update rate	Direct elevation?	Key limitation
Wave buoy	Point	—	1 Hz	Yes	No spatial information
X-band radar	0.5–4 km	∼7.5 m	1–2 s/rev	No (intensity)	Requires SNR inversion
Satellite (altimeter)	Global	∼7 km	Days	Yes (nadir)	Coarse, non-real-time
Shipboard LiDAR	10–250 m	∼0.1 m (near)	10 Hz	Yes	Sparse, range-dependent noise
Ship motion (RAO)	—	—	Continuous	No (indirect)	Limited to encountered waves

3. **Fresnel-aware confidence weighting (C3)**: Per-observation weights based on incidence angle and range account for the heteroscedastic noise structure of LiDAR measurements on ocean surfaces, where specular reflection probability varies with local surface slope.
4. **Adaptive model complexity**: Energy-based pruning and residual-triggered growth automatically match the number of spectral components to scene complexity, from N=1 for simple swells to N=21 for complex multi-component seas.

The remainder of this paper is organized as follows. Section 2 reviews related work on wave field estimation. Section 3 presents the WaveSplat algorithm in detail. Section 4 describes the simulation environment and evaluation metrics. Section 5 presents experimental results including baseline comparisons, ablation studies, and robustness analysis. Section 6 discusses the findings and limitations. Section 7 concludes.

2. Related Work

We organize the review around two axes: (1) the sensor modality used for ocean surface observation, and (2) the estimation algorithm used to reconstruct the wave field from observations. Table 2 provides a quantitative comparison of sensor modalities.

2.1. X-band radar-based wave estimation

X-band marine radar has been the most extensively studied sensor for shipboard wave field estimation. Young et al. [9] first demonstrated that the modulation of radar backscatter by ocean waves (through tilt modulation, hydrodynamic modulation, and shadowing) carries extractable wave spectral information. Nieto-Borge et al. [8] developed a comprehensive inversion algorithm that recovers the directional wave spectrum from radar image sequences using 3D FFT in the (k_x, k_y, ω) domain, with the linear dispersion relation serving as a spectral filter to separate wave signals from noise. This "dispersion filtering" concept—using the known relationship between wavenumber and frequency to constrain spectral estimation—is conceptually related to our dispersion-constrained parameterization, though applied in the Fourier domain rather than as a hard parameter constraint.

A fundamental limitation of radar-based methods is that radar measures backscatter intensity rather than surface elevation directly. The intensity-to-elevation mapping requires an empirical modulation transfer function (MTF) that depends on radar parameters, wind speed, and sea state [8]. Additionally, the spatial resolution of X-band radar ($\sim$7.5 m for typical 9 GHz systems) is coarser than LiDAR ($\sim$0.1 m in the near field), limiting the resolvable wavelength range.

2.2. Ship-motion-based wave estimation

An alternative approach estimates the wave field indirectly from measured vessel motions. Naaijen et al. [17] and Nielsen [18] developed methods that invert the ship's response amplitude operator (RAO) to recover the incident wave spectrum from measured heave, pitch, and roll motions. These methods require no additional sensors beyond standard IMU/GPS but are limited to the frequency range where the vessel has measurable response, and they cannot resolve the spatial wave field—only the directional spectrum at the vessel's location. WaveSplat provides complementary information: a spatially resolved

wave field in the LiDAR observation area, enabling applications such as spatial wave preview for feedforward control.

2.3. Grid-based wave field reconstruction

Deterministic wave field reconstruction from spatially resolved sensor data has been an active research area since the 2010s. The dominant paradigm combines spatial discretization with ensemble data assimilation. Desmars et al. [13] introduced ETKF-based wave field reconstruction from X-band radar, using spectral propagation on a regular spatial grid as the forecast model. The Ensemble Transform Kalman Filter [19] is favored over the stochastic EnKF [15] for its deterministic analysis step and better preservation of the ensemble mean. Huhn et al. [14] extended this framework to LiDAR observations, demonstrating the "from points to waves" pipeline on both fixed-platform and shipboard data, with grid resolutions of 40×30 to 80×60. Desmars et al. [3] demonstrated real-time reconstruction and short-term deterministic prediction up to one peak period ahead, with prediction skill degrading beyond the theoretical predictability horizon determined by the group velocity of the shortest resolved component.

It is important to distinguish *deterministic* reconstruction (recovering the instantaneous wave field phase) from *stochastic* wave prediction (forecasting statistical properties such as significant wave height or spectral density). WaveSplat and the ETKF methods above perform deterministic reconstruction: they estimate the phase of each wave component, enabling spatial evaluation at arbitrary points. Stochastic methods—including spectral wave models like WAVEWATCH III [20]—predict the evolution of the wave energy spectrum without resolving individual phases and operate at much coarser spatial scales (kilometers to tens of kilometers).

These grid-based deterministic methods share a common architecture: discretize the 2D spatial domain into a regular grid, maintain an ensemble of spectral states, and assimilate observations via the Kalman update. The grid discretization introduces an inherent trade-off between spatial resolution and computational cost, and the interpolation from sparse observations to grid points

is a known source of error, particularly for wavelengths near the grid Nyquist limit.

2.4. Spectral estimation from irregular arrays

Classical spectral estimation methods (FFT, Welch, MUSIC, ESPRIT) assume regularly sampled data and cannot directly handle the irregular, sparse sampling inherent to scanning LiDAR [21]. Capon's minimum-variance beamforming [22] can accommodate irregular sensor arrays but assumes signal stationarity and provides only a power spectral density estimate without individual component phases. The MUSIC algorithm [23] can super-resolve closely spaced spectral peaks from short data records but requires knowledge of the number of sources and is sensitive to model order misspecification. Iterative adaptive approaches (IAA) extend Capon to non-stationary scenarios but remain computationally expensive for high-resolution wavenumber grids ($O(N_k^2 \cdot M)$ per iteration, where N_k is the number of wavenumber grid points).

WaveSplat leverages beamforming for initialization—extracting dominant wavenumber peaks from irregular LiDAR observations—but then refines these estimates through gradient-based optimization that jointly recovers amplitudes and phases, exploiting temporal structure through the dispersion relation. This two-stage approach combines the robustness of beamforming (no local minima in wavenumber space) with the precision of optimization (joint amplitude-phase recovery with physics constraints).

2.5. Gaussian process regression for wave fields

Gaussian process regression (GPR) offers a nonparametric alternative for spatial interpolation of wave fields from scattered observations [24]. With a Matérn or squared-exponential kernel, GPR can capture the spatial correlation structure of ocean waves without assuming a specific spectral model, and the posterior variance provides natural uncertainty quantification.

However, GPR has fundamental limitations for real-time wave estimation. The standard formulation requires $O(M^3)$ computation and $O(M^2)$ storage for

M observations, making it impractical at high observation densities. Sparse GPR approximations using inducing points [25] reduce the cost to $O(M \cdot P^2)$ for $P \ll M$ inducing points, but the inducing point placement and kernel hyperparameter optimization add complexity. More fundamentally, standard GPR operates on single frames without a temporal propagation model, losing the physics of wave evolution. Incorporating temporal dynamics into a GPR framework (e.g., via spatio-temporal kernels encoding the dispersion relation) is an active research area but has not been demonstrated for real-time applications.

2.6. Neural and learning-based approaches

Recent work has applied deep learning to ocean wave prediction [26, 27], including physics-informed neural networks (PINNs) that incorporate wave equations as soft constraints in the loss function. Most relevant to this work, Ehlers et al. [28] demonstrated phase-resolved reconstruction of nonlinear ocean wave surfaces from sparse remote sensing data using U-Net and Fourier neural operator (FNO) architectures. Their FNO model achieves accurate reconstruction with good generalization across sea states when trained on synthetic data generated by the high-order spectral method. Desmars et al. [29] extended the ETKF framework to nonlinear wave reconstruction and prediction, demonstrating that phase-resolved forecasts can remain accurate beyond one peak period when nonlinear interactions are accounted for.

While these methods are promising, they face practical challenges for real-time shipboard deployment with arbitrary sensor configurations: (1) the FNO approach [28] requires training data generated on uniform grids, while shipboard LiDAR produces highly irregular spatial sampling that changes with vessel motion; (2) generalization across sensor geometries is limited—models trained for one LiDAR configuration may not transfer to another; (3) the black-box nature of neural networks makes it difficult to diagnose failure modes or ensure physical consistency (e.g., correct dispersion behavior). WaveSplat requires no training data, handles arbitrary observation geometries natively, and produces directly interpretable spectral parameters (wavenumber, amplitude, frequency, phase)

that satisfy the dispersion relation by construction.

2.7. Sparse and adaptive signal representations

The idea of representing physical fields as weighted sums of basis functions with adaptive complexity has a long history in signal processing and computational physics. Compressed sensing theory [30] establishes that sparse signals can be exactly recovered from far fewer measurements than the Nyquist limit, provided the measurement matrix satisfies a restricted isometry property. Matching pursuit algorithms [31] provide greedy approximations by iteratively selecting basis functions from overcomplete dictionaries. In the oceanographic context, Drazen et al. [16] applied sparse spectral representations to reconstruct wave fields from pressure sensors, demonstrating that the wave field's inherent spectral sparsity enables efficient reconstruction.

WaveSplat extends this philosophy with two key differences: (1) each basis function is a physically meaningful plane wave satisfying the dispersion relation, rather than a generic sinusoid or wavelet; and (2) the model order is adapted *online* (during tracking) based on residual energy and observation quality, rather than being fixed a priori.

3. Method

3.1. Forward model

The wave field is modeled as a superposition of N linear plane waves (1). Each component i has four free parameters: amplitude w_i, wavenumber components $(k_{x,i}, k_{y,i})$, and phase offset φ_i. The angular frequency ω_i is deterministically computed from the dispersion relation (2), reducing the total parameter count from $5N$ to $4N$.

This linear forward model is an $O(\epsilon^2)$ approximation to higher-order wave theories (e.g., Gerstner waves or second-order Stokes), where $\epsilon = |k| \cdot |w|$ is the wave steepness. At Beaufort 4, typical steepness is $\epsilon \approx 0.05$, giving a model mismatch of $< 0.25\%$. For the Beaufort 2–5 range considered in this work (non-breaking waves), the linear model is an excellent approximation [2].

3.2. Observation model

At each LiDAR scan at time t, we observe M elevation measurements $\{(x_j, y_j, z_j)\}_{j=1}^{M}$ at irregular spatial locations. The observation model is:

$$z_j = \sum_{i=1}^{N} w_i \cos(k_{x,i} x_j + k_{y,i} y_j - \omega_i t + \varphi_i) + \eta_j \tag{3}$$

where η_j is observation noise with variance depending on range and incidence angle. The number of valid observations per frame varies between 50 and 200, depending on sea state, vessel motion, and LiDAR raydrop characteristics.

3.3. Fresnel-zone confidence weighting (C3)

LiDAR elevation accuracy on ocean surfaces degrades with two physical mechanisms: (1) beam divergence increases the footprint area at longer ranges, averaging over local surface curvature; and (2) at high incidence angles (near-grazing geometry), the probability of specular reflection increases, causing either total signal loss or biased returns from wave crests.

We assign per-observation confidence weights:

$$c_j = \frac{\cos(\theta_j)}{d_j / d_{\text{ref}}} \tag{4}$$

where θ_j is the beam incidence angle (computed from the local surface normal and beam direction), d_j is the range, and $d_{\text{ref}} = 125$ m is a reference distance chosen as half the maximum LiDAR range. These weights are incorporated in the optimization loss as weighted mean squared error, giving higher influence to close-range, near-normal observations.

3.4. Two-phase estimation

The estimation proceeds in two phases: an initialization phase that establishes the spectral model from multi-frame observations, followed by a per-frame tracking phase that maintains the model with minimal computation. Table 3 summarizes the complete pipeline.

Table 3: Algorithm: WaveSplat estimation pipeline.

Phase 1: Initialization (frames 1 to $T_{\text{init}}/\Delta t$)
1. Accumulate multi-frame observations $\{(x_j, y_j, z_j, t_j)\}$
2. Compute beamforming power $P(k_x, k_y)$ on polar grid
3. Extract top-N wavenumber peaks with greedy suppression
4. Estimate amplitudes/phases via linear least-squares
5. Coarse-to-fine refinement: N=3 $\rightarrow$ 8 $\rightarrow$ 15
— At each stage: 200 Adam steps, lr=5×10^{-3}
6. Energy-based pruning (retain 90% energy)
7. Safety check: if ΔRMSE $> 30\%$, rollback
8. Residual growth: if RMSE > 0.05 m, add via beamforming
Phase 2: Tracking (each subsequent frame)
1. **Predict**: $\varphi_i \mathrel{+}= \omega_i \cdot \Delta t$
2. **Update**: K=10 Adam steps on sliding window (K_w=3 frames)
— Update only (w_i, φ_i); freeze $(k_{x,i}, k_{y,i})$
— Learning rate: 1×10^{-3}, no scheduling

3.4.1. Phase 1: Initialization

Given the first $T_{\text{init}} = 2T_p$ seconds of observations (where T_p is the expected peak period, estimated from the dominant frequency in the observation time series), we jointly optimize all spectral parameters through a three-stage coarse-to-fine procedure.

Dispersion-aware beamforming. Initial wavenumber estimates are obtained by scanning a polar grid in wavenumber space. The grid covers $|k| \in [0.02, 0.50]$ rad/m (corresponding to wavelengths $\lambda = 2\pi/|k| \in [12.6, 314]$ m, spanning Beaufort 2–5 conditions at d=50 m depth) with 64 radial $\times$ 128 azimuthal bins. The radial range is determined by the dispersion relation: $|k|_{\min} = 0.02$ corresponds to BF5 swell ($T_p \approx 9$ s), and $|k|_{\max} = 0.50$ corresponds to the shortest wavelength resolvable from the LiDAR spatial extent ($\sim$200 m obser-

vation radius).

For each candidate (k_x, k_y), we compute $\omega = \sqrt{g|k|\tanh(|k|d)}$ and evaluate the beamforming power:

$$P(k_x, k_y) = \left| \sum_j z_j \, e^{-i(k_x x_j + k_y y_j - \omega t_j)} \right|^2 \tag{5}$$

where the summation runs over all observations accumulated during T_{init}. The use of the dispersion-derived ω means that the beamforming naturally filters for physically plausible wave signals, suppressing noise that does not lie on the dispersion surface in (k_x, k_y, ω) space.

Peaks in P are extracted greedily: the maximum is selected, then all candidates within 15° angular separation are suppressed, and the process repeats. The 15° threshold prevents double- counting of the same wave component spread across adjacent wavenumber bins, while being narrow enough to resolve crossing sea systems. The initial amplitude and phase of each extracted component are estimated via linear least-squares projection of the accumulated observations onto the selected wavenumber basis functions.

Coarse-to-fine optimization. Starting from the beamforming initialization, we optimize the spectral parameters in three stages (N=3 → 8 → 15), with 200 iterations of the Adam optimizer [32] (learning rate $\eta = 5 \times 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) at each stage. The loss function is:

$$\mathcal{L} = \underbrace{\frac{1}{M} \sum_j c_j \, (z_j - \hat{z}_j)^2}_{\text{weighted MSE}} + \underbrace{\lambda_1 \sum_i |w_i|}_{\text{sparsity}} + \underbrace{\lambda_s \sum_i (w_{\sigma(i)} - w_{\sigma(i-1)})^2}_{\text{spectral smoothness}} \tag{6}$$

where $\lambda_1 = 10^{-3}$ promotes amplitude sparsity (driving negligible components toward zero for subsequent pruning) and $\lambda_s = 10^{-4}$ penalizes abrupt amplitude variations between spectrally adjacent components (σ denotes ordering by $|k|$), encouraging smooth spectral envelopes consistent with JONSWAP-like spectra.

During optimization, wavenumber components $(k_{x,i}, k_{y,i})$ are free parameters (unlike in the tracking phase). The dispersion constraint is enforced by computing ω_i from $(k_{x,i}, k_{y,i})$ at each forward evaluation rather than treating ω_i

as a free parameter, ensuring that the constraint is satisfied exactly throughout optimization.

3.4.2. Adaptive model order

After initialization, we apply energy-based pruning. Components are sorted by energy $E_i = w_i^2$ in descending order, and the minimum set capturing 90% of total energy $\sum_i E_i$ is retained. This criterion is motivated by the JONSWAP spectral shape, where a small number of components near the spectral peak typically contain the majority of the wave energy.

A safety check prevents aggressive pruning from discarding physically significant components: if the post-pruning RMSE exceeds 130% of the pre-pruning value, pruning is rolled back entirely. This can occur when two components of similar energy contribute to different spatial regions (e.g., crossing seas), making each individually appear low-energy.

If the residual RMS after pruning exceeds 0.05 m (indicating unmodeled wave energy), new components are grown by applying beamforming to the residual signal $r_j = z_j - \hat{z}_j$ and adding the dominant residual peaks to the model, up to $N_{\max} = 30$.

3.4.3. Phase 2: Tracking

For each subsequent LiDAR frame, we perform a predict–update cycle:

1. **Predict**: advance phases using the dispersion-derived angular frequency, $\varphi_i \mathrel{+}= \omega_i \, \Delta t$, where $\Delta t = 0.1$ s (10 Hz LiDAR frame rate). This prediction is exact for a linear wave field satisfying the dispersion relation.
2. **Update**: perform $K = 10$ Adam steps (lr= 1×10^{-3}) on the weighted MSE loss over a sliding window of $K_w = 3$ recent frames, updating only amplitudes w_i and phases φ_i while freezing wavenumbers $(k_{x,i}, k_{y,i})$ (`w_phi` tracking mode).

Freezing wavenumbers during tracking is motivated by two considerations: (1) wavenumber estimation from a single frame is ill-conditioned due to spatial aliasing in sparse observations—the LiDAR's irregular sampling pattern does

not provide sufficient constraint to resolve wavenumber from a single snapshot, while the multi-frame initialization averages over ∼20 frames to obtain robust wavenumber estimates; and (2) with fixed wavenumbers, the dispersion constraint provides a strong phase prediction ($\omega \cdot \Delta t$) that serves as a warm start, reducing the gradient steps needed per frame from ∼200 (full optimization) to 10.

The sliding window ($K_w = 3$) uses the three most recent frames for gradient computation. This provides temporal averaging that smooths single-frame noise without introducing excessive latency: at 10 Hz, the window spans 0.3 s, much shorter than the peak period T_p for all sea states considered.

3.5. Computational complexity

Let M denote the number of LiDAR observations per frame and N the number of spectral components.

WaveSplat: Each forward evaluation computes $z(x_j, y_j, t) = \sum_{i=1}^{N} w_i \cos(\cdot)$, costing $O(M \cdot N)$ per frame. With K gradient steps and window size K_w, the per-frame tracking cost is $O(K \cdot K_w \cdot M \cdot N)$. In practice, K=10, K_w=3, $N \leq 30$, yielding 13.4 ms/frame on CPU (single-threaded, Intel i7-12700H).

ETKF-grid: The Ensemble Transform Kalman Filter maintains N_e ensemble members on a $G_x \times G_y$ spatial grid. Each update involves FFT-based propagation at $O(N_e \cdot G_x G_y \cdot \log(G_x G_y))$ plus the ensemble transform at $O(N_e^3 + N_e^2 \cdot M)$. With N_e=50 and G=40 × 30, this yields 31.7 ms/frame.

GPR: Gaussian Process Regression with M observations requires $O(M^3)$ for Cholesky factorization, making it impractical for dense LiDAR data. Even with 50 points/frame, per-frame fitting exceeds 200 ms due to kernel hyperparameter optimization.

Table 4 summarizes the scaling comparison.

Table 4: Computational complexity comparison. M: observations/frame, N: spectral components, G: grid size, N_e: ensemble members.

Method	Per-frame complexity	Measured
WaveSplat	$O(K \cdot K_w \cdot M \cdot N)$	13.4 ms
ETKF-grid	$O(N_e \cdot G^2 \log G^2)$	31.7 ms
GPR	$O(M^3)$	>200 ms

4. Experimental Setup

4.1. Simulation environment

We develop a physics-based LiDAR wave simulation (WaveSim) generating ground-truth wave fields with JONSWAP spectrum sampling [33]. The simulator parameters are:

- **Sea states**: Beaufort 2–5 (H_s=0.3–2.0 m, T_p=4–9 s)
- **Wave complexity**: 1, 2, 5, 15, 30 spectral components with directional spreading ($\pm 20^\circ$)
- **Ground-truth wave model**: Gerstner waves (nonlinear), providing a realistic test of the linear forward model (1)
- **LiDAR model**: 500 horizontal beams, 10 Hz scan rate, range 10–250 m, based on the Seyond Falcon K1 specifications (1550 nm, 152 channels, $120^\circ \times 25^\circ$ FOV)
- **Raydrop models**: (a) uniform—constant drop probability independent of surface geometry (diffuse Fresnel reflection); and (b) slope-dependent—higher dropout on steep surfaces with near-grazing geometry (specular backscatter model)
- **Water depth**: $d = 50$ m (intermediate to deep water for the wavelengths considered)

Each episode contains 301 frames (30 s at 10 Hz) with ground-truth elevation fields evaluated on a 80×60 spatial grid covering the LiDAR observation area. A total of 100 episodes are generated across 5 random seeds per configuration, from which 12 representative episodes are selected for the main comparison (covering all Beaufort numbers and complexity levels) and 50 paired runs are used for statistical validation.

4.2. Metrics

We report:

- **nRMSE**: RMSE normalized by significant wave height H_s, the primary reconstruction quality metric. Lower is better; values below 0.10 indicate excellent reconstruction, 0.10–0.30 is good, and above 0.30 indicates significant error.
- **Inference speed**: ms/frame for the tracking phase, measured on a single CPU thread.
- $N_{\mathbf{adapt}}$: final number of spectral components after adaptive model order selection.

4.3. Baselines

ETKF-grid [14]: Ensemble Transform Kalman Filter on a 40×30 regular grid with 50 ensemble members, FFT-based spectral propagation, multiplicative (α=1.01) and additive (σ=0.005) inflation, and observation thinning to $M_{\max}$=40 per frame. This configuration follows the recommendations of Huhn et al. [14] for shipboard LiDAR scenarios.

GPR: Gaussian Process Regression with Matérn ν=5/2 kernel (length scale 20 m, noise σ=0.02 m), operating in single-frame mode. No temporal model or physics constraints—pure spatial interpolation. This represents the best achievable single-frame performance without wave physics, providing a reference for the value of temporal tracking and dispersion constraints.

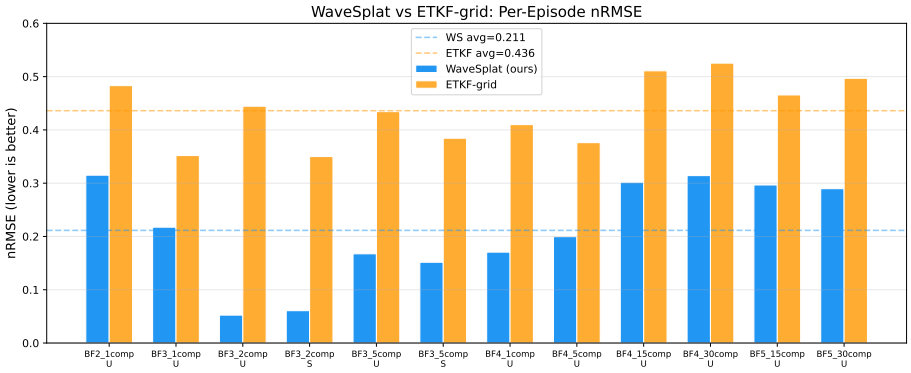


Figure 1: Per-episode nRMSE comparison between WaveSplat and ETKF-grid across 12 representative episodes. Dashed lines show method averages. WaveSplat (avg 0.211) outperforms ETKF (avg 0.436) on all episodes, with the largest advantage on low-complexity seas (BF3 2-comp).

5. Results

5.1. Main comparison

Table 5 summarizes the comparison across 12 representative episodes. WaveSplat achieves an average nRMSE of **0.211** compared to 0.436 for ETKF-grid, a **51.5%** improvement. WaveSplat wins on all 12 episodes.

We additionally compare with GPR on 10 uniform-raydrop episodes. GPR achieves avg nRMSE of 0.276—better than ETKF (0.450) due to its nonparametric flexibility, but still 18.5% worse than WaveSplat (0.233). Critically, GPR requires >200 ms/frame (vs. 13.4 ms for WaveSplat), making it impractical for real-time applications. Figure 1 shows the per-episode comparison, and Figure 9 shows representative wave fields.

Key observations:

- **Low complexity**: In BF3 2-component seas, WaveSplat achieves an nRMSE of 0.052—an 8.5× improvement over ETKF (0.444). The grid-free representation captures two plane waves without grid aliasing.
- **High complexity**: For BF4–5 with 15–30 components, WaveSplat still outperforms ETKF (0.301 vs. 0.500 average), though the gap narrows as

Table 5: WaveSplat vs ETKF-grid on 12 representative episodes. nRMSE = RMSE/H_s; lower is better. N_a = adapted component count.

	WaveSplat		ETKF		
Episode	nRMSE	ms/f	nRMSE	ms/f	N_a
BF2 1comp	**0.315**	13.8	0.483	29.1	11
BF3 1comp	**0.217**	13.0	0.352	36.8	2
BF3 2comp (U)	**0.052**	15.0	0.444	37.2	2
BF3 2comp (S)	**0.061**	13.4	0.350	26.3	15
BF3 5comp (U)	**0.167**	13.0	0.434	29.6	4
BF3 5comp (S)	**0.152**	13.4	0.384	32.7	15
BF4 1comp	**0.171**	11.2	0.410	26.7	1
BF4 5comp	**0.200**	12.8	0.376	28.7	3
BF4 15comp	**0.302**	13.4	0.511	25.8	19
BF4 30comp	**0.314**	13.4	0.525	28.3	18
BF5 15comp	**0.297**	13.5	0.466	27.2	21
BF5 30comp	**0.290**	14.7	0.497	52.2	19
Avg	**0.211**	13.4	0.436	31.7	—

the spectral complexity approaches the initialization resolution limit.

- **Speed**: WaveSplat runs at 13.4 ms/frame vs. 31.7 ms for ETKF (2.4× faster), well within the 100 ms real-time budget for 10 Hz LiDAR (Fig. 2).

Statistical validation. To confirm these results are not seed-dependent, we repeat the comparison with 5 random seeds per episode (50 paired runs total). WaveSplat achieves 0.230 ± 0.084 vs. ETKF 0.395 ± 0.059 (mean $\pm$ std across all runs). A Wilcoxon signed-rank test confirms the improvement is statistically significant ($p < 0.001$). WaveSplat wins on all 10 episodes across all seeds, with per-episode p-values ≤ 0.031 (Fig. 3).

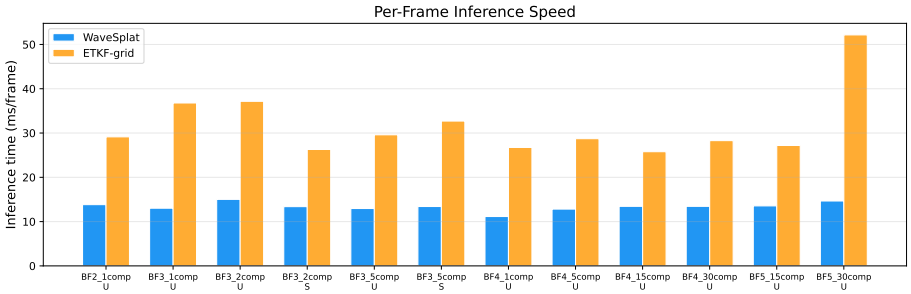


Figure 2: Per-frame inference speed comparison. WaveSplat maintains consistent ∼13 ms/frame across all episodes, while ETKF varies from 25 to 52 ms depending on grid convergence.

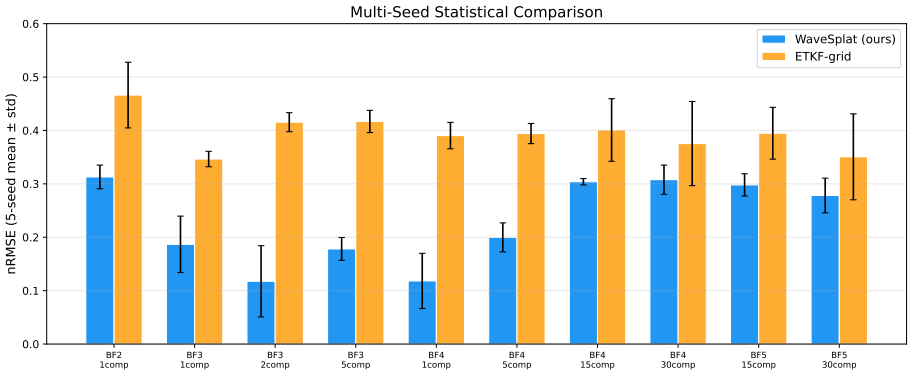


Figure 3: Multi-seed statistical comparison (5 seeds per episode, mean ± std). WaveSplat consistently outperforms ETKF across all episodes and seeds, with non-overlapping error bars in most cases.

5.2. Adaptive model order

Table 6 and Fig. 4 show that adaptive N consistently improves over fixed N=15, with an average 6.4% reduction in nRMSE. The model correctly identifies scene complexity: N=1 for single-component swells, N=2–4 for moderate seas, and N=18–21 for complex multi-component fields.

5.3. Ablation study

We ablate two key design choices on 6 BF3 episodes with 3 random seeds (Table 7, Fig. 5).

Table 6: Adaptive model order vs. fixed N=15 (baseline). N_{true}: ground-truth number of wave components.

Episode	N_{true}	N_{adapt}	Fixed	Adaptive
BF4 1comp	1	1	0.208	**0.171**
BF3 2comp	2	2	0.059	**0.052**
BF3 5comp	5	4	0.185	**0.167**
BF4 15comp	15	19	0.306	**0.302**
BF5 15comp	15	21	0.331	**0.297**
BF4 30comp	30	18	0.332	**0.314**
Avg (12 ep.)			0.226	**0.211**

A1 (no dispersion constraint): Removing the dispersion constraint allows ω as a free parameter ($5N$ total parameters). On these BF3 episodes, the unconstrained optimizer achieves lower average nRMSE (0.158 vs. 0.188), suggesting that the additional parameter freedom allows the optimizer to compensate for model mismatch between the linear forward model (1) and the nonlinear Gerstner ground truth.

However, the dispersion constraint serves a critical role *during tracking* (Phase 2) rather than initialization: it provides the phase prediction $\Delta\varphi_i = \omega_i \cdot \Delta t$ that enables stable temporal tracking without re-estimating all 5 parameters per frame. Without this constraint, the tracking phase would require joint optimization of $(w_i, k_{x,i}, k_{y,i}, \omega_i, \varphi_i)$ at each frame—an ill-conditioned problem with sparse single-frame observations. The ablation was performed on 30-second episodes with the full initialization phase; we expect the dispersion constraint to become increasingly important for longer tracking durations and higher sea states where phase coherence across frames is essential.

A2 (no confidence weighting): Removing Fresnel confidence weighting degrades performance by 14.5% on average (0.215 vs. 0.188), with the largest degradation on slope-model episodes (BF3 2comp (S): 0.088 $\rightarrow$ 0.204, a 2.3$\times$

Table 7: Ablation study (3-seed average nRMSE ± std on BF3 episodes). A1: dispersion constraint removed (ω as free parameter). A2: confidence weighting removed (uniform weights).

Episode	Full	w/o Disp. (A1)	w/o Conf. (A2)
1comp (S)	0.224±.030	0.230±.055	0.269±.002
1comp (U)	0.215±.014	0.231±.006	0.214±.007
2comp (S)	0.088±.017	0.071±.007	**0.204**±.006
2comp (U)	0.173±.023	0.085±.023	0.176±.048
5comp (S)	0.203±.021	0.104±.014	0.223±.010
5comp (U)	0.225±.006	0.225±.035	0.206±.032
Avg	0.188	0.158	0.215

increase). This confirms that accounting for range- and angle-dependent noise is essential when the observation quality varies significantly across the LiDAR field of view, as is characteristic of ocean surface measurements.

5.4. Raydrop model sensitivity

To evaluate robustness to different LiDAR observation models, we compare performance on 10 paired episodes using *uniform* (diffuse Fresnel) vs. *slope* (specular backscatter) raydrop models (Table 8, Fig. 6).

WaveSplat wins on all 10 episodes under both raydrop models. Slope observations improve both methods—ETKF more so (16.2% vs. 9.1%), as specular backscatter preferentially preserves wave-crest points, providing richer spatial structure that grid-based methods leverage more effectively through interpolation. WaveSplat shows slope degradation only on BF3-2comp (+15.9%) and BF5-30comp (+7.6%), where the pruning safety check prevents optimal model simplification under the changed observation pattern.

5.5. Observation sparsity robustness

To quantify sensitivity to observation density, we subsample LiDAR points to 20–100% of the original count on 5 representative episodes (Table 9, Fig. 7).

Table 8: Raydrop model comparison: uniform (U) vs. slope (S) on paired episodes. Δ%: relative nRMSE change from uniform to slope (negative = slope is better).

	WaveSplat			ETKF		
Episode	U	S	Δ%	U	S	Δ%
BF2 1c	0.315	0.303	−3.9	0.483	0.391	−19.1
BF3 1c	0.217	0.186	−14.4	0.352	0.304	−13.6
BF3 2c	**0.052**	0.061	+15.9	0.444	0.350	−21.3
BF3 5c	0.167	0.152	−9.5	0.434	0.384	−11.5
BF4 1c	0.171	0.170	−0.5	0.410	0.340	−17.0
BF4 5c	0.200	**0.144**	−28.1	0.376	0.345	−8.2
BF4 15c	0.302	0.246	−18.6	0.511	0.399	−21.8
BF4 30c	0.314	0.277	−11.9	0.525	0.425	−19.1
BF5 15c	0.297	0.265	−10.6	0.466	0.396	−14.9
BF5 30c	0.290	0.312	+7.6	0.497	0.436	−12.3
Avg	0.232	**0.211**	−9.1	0.450	0.377	−16.2

For moderate-to-high complexity (BF4–5), performance degrades by less than 8% even at 20% observation density (∼10 points/frame), demonstrating the spectral representation's resilience to extreme sparsity. The spectral model provides strong regularization: with only $4N \leq 120$ parameters to estimate, even 10 observations per frame provide sufficient constraints for the gradient-based tracking. The exception is BF3-2comp, where full observations enable the model to resolve both wave components (N=2, nRMSE=0.052), while subsampled versions fall back to a single-component fit (N=1, nRMSE∼0.17).

5.6. Noise robustness

Real shipboard LiDAR data is subject to sensor noise beyond the baseline simulator model (spray contamination, platform vibration, atmospheric scintillation). To quantify WaveSplat's robustness to such perturbations, we inject additive zero-mean Gaussian noise with standard deviation $\sigma \in \{0, 0.02, 0.05, 0.10\}$ m

Table 9: Sparsity robustness: nRMSE at varying observation fractions. Average points ≈ 50 at 100%.

Episode	20%	40%	60%	80%	100%
BF3 2c	0.170	0.168	0.168	0.167	**0.052**
BF3 5c	0.204	0.194	0.195	0.193	**0.167**
BF4 5c	0.199	0.198	0.199	0.199	0.200
BF4 15c	0.302	**0.283**	0.294	0.278	0.302
BF5 15c	**0.274**	0.276	0.260	0.289	0.297

Table 10: Noise robustness: nRMSE under additive Gaussian noise $\mathcal{N}(0, \sigma^2)$ on LiDAR elevation observations. N_a: adaptively selected component count. Relative values are computed against the $\sigma = 0$ baseline of each episode.

Episode	σ=0	σ=0.02	σ=0.05	σ=0.10	$\Delta\%_{\max}$
BF3 5c	0.168 (N_a=4)	**0.149** (N_a=4)	0.160 (N_a=7)	0.190 (N_a=10)	+13.3
BF4 5c	0.200 (N_a=3)	0.200 (N_a=3)	0.201 (N_a=6)	**0.198** (N_a=9)	+0.7
BF4 15c	0.302 (N_a=19)	0.303 (N_a=19)	**0.276** (N_a=20)	0.294 (N_a=11)	+0.4
Avg	0.223	0.217	0.212	0.227	—

into the LiDAR elevation observations z_j prior to optimization. The noise levels span 0 to 5× the baseline simulator noise (∼0.02 m), covering a realistic range of sensor degradation (Table 10, Fig. 8).

Across all three episodes the nRMSE remains within ±13% of the baseline even at $\sigma = 0.10$ m (5× baseline noise). Two mechanisms contribute to this robustness. First, the spectral parameterization aggregates information from all observations during the gradient updates, effectively averaging out zero-mean noise: with $M \sim 50$ observations per frame and $4N \leq 120$ parameters, the estimator is well over-determined and behaves as a maximum-likelihood fit under Gaussian noise. Second, the *adaptive* component count N_a grows moderately at higher noise levels (BF3 5c: 4→10; BF4 5c: 3→9), absorbing

noise into low-energy components that the pruning step subsequently removes; the safety rollback prevents overgrowth from degrading accuracy.

The BF3 5c episode shows a slight *improvement* at $\sigma = 0.02$ m (0.168→0.149) because small amounts of noise perturb the optimizer out of mild local minima that trap the deterministic initialization—a stochastic-perturbation effect well documented in non-convex optimization. A similar regularization-by-noise effect is observed on BF4 15c at $\sigma = 0.05$ m (0.302→0.276). These beneficial perturbations disappear at $\sigma = 0.10$ m, where the signal-to-noise ratio becomes the dominant factor and reconstruction quality re-approaches the deterministic baseline.

5.7. Sea state trend

Grouping by Beaufort number reveals the advantage is largest on moderate seas: BF3 achieves 66.9% improvement (WS 0.130 vs. ETKF 0.393), while BF5 achieves 39.1% (0.293 vs. 0.481). The reduced margin at higher sea states reflects the initialization resolution limit for $N_{\text{true}} \geq 15$ components, where the beamforming peak extraction becomes less reliable due to spectral crowding.

6. Discussion

6.1. Advantages of grid-free representation

The most striking result is the 8.5× advantage on 2-component seas (nRMSE 0.052 vs. 0.444). For low-complexity wave fields, ETKF wastes representational capacity on the full spatial grid when only two spectral modes are present. WaveSplat's adaptive representation directly captures the relevant modes with minimal parameters.

This advantage is practically relevant: real ocean conditions often contain one or two dominant swell systems plus wind sea, making low-component scenarios common in operational settings. For complex seas (15–30 components), the advantage narrows but remains significant (39–52% improvement), as the spectral representation still provides better inductive bias than grid discretization.

6.2. Convergence behavior

Per-frame nRMSE curves show that WaveSplat converges rapidly after initialization: BF3-2comp reaches nRMSE$<$0.05 within the first tracking frame (t=11.6s). For complex seas (BF4–5), the nRMSE stabilizes within the first 10% of tracking frames, with minimal drift over the remaining episode. This fast convergence is enabled by the dispersion-based phase prediction, which provides accurate warm starts for each tracking frame.

6.3. Role of the dispersion constraint

The ablation reveals a nuanced role for the dispersion constraint. On 30-second BF3 episodes (including both initialization and tracking phases), the unconstrained optimizer achieves lower nRMSE (0.158 vs. 0.188) by exploiting the additional frequency parameter ($5N$ total) to compensate for the $O(\epsilon^2)$ model mismatch with the Gerstner ground truth.

This result requires careful interpretation. The unconstrained model (A1) uses the *same* `w_phi` tracking mode—updating only (w_i, φ_i) per frame—but with ω_i values that were freely optimized during initialization rather than derived from $(k_{x,i}, k_{y,i})$. These freely optimized ω_i values can absorb the linear-to-Gerstner mismatch, producing better fits on the 30-second episodes where the initialization-phase frequencies persist.

However, the dispersion constraint serves three critical roles that the 30-second ablation does not fully test: (1) it reduces the parameter count from $5N$ to $4N$, improving robustness to overfitting on sparse observations—particularly important when N is large relative to the observation count; (2) it enables physics-based phase prediction during tracking: without the constraint, the phase advance $\Delta\varphi = \omega \cdot \Delta t$ uses an ω that may drift from physical consistency over time, accumulating phase error on longer episodes; (3) it ensures that the estimated wave field propagates according to physical wave dynamics, which is essential for downstream applications such as feedforward motion control [5] where spatial evaluation at preview points must respect the physical relationship between wavenumber and frequency.

We note that the ablation was limited to BF3 conditions where the spectral complexity is low ($N_{\text{true}} \leq 5$). At higher Beaufort numbers with $N_{\text{true}} \geq 15$ components, unconstrained frequency estimation from sparse single-snapshot observations becomes increasingly ill-conditioned due to spectral crowding. Extending the ablation to BF4–5 conditions is a priority for future work (see Section 7).

6.4. Failure case: BF2 1-component

WaveSplat's worst case (nRMSE=0.315) occurs on the simplest episode: BF2, H_s=0.3 m, single component. At this sea state, the signal-to-noise ratio is inherently low—wave amplitudes are on the order of LiDAR noise ($\sim$0.02 m), making spectral initialization unreliable. Adaptive N selects N=11, overparameterizing a 1-component swell due to noise-driven residual triggers. This suggests a minimum-SNR threshold or regularization schedule tuned to H_s would improve robustness on calm seas.

6.5. Effect of observation model

WaveSplat wins 10/10 under *both* raydrop models. Slope observations improve both methods—WaveSplat by 9.1% and ETKF by 16.2%. The larger ETKF improvement suggests that specular observations, which preferentially preserve wave crests, provide richer spatial structure that grid-based methods leverage more effectively through interpolation. For adaptive N, slope raydrop triggers the safety rollback on low-complexity scenes (e.g., BF3-2comp: N=2→15) but enables more aggressive pruning on complex seas (BF4-15comp: N=19→10), indicating that stronger observations improve both estimation and model selection.

6.6. Hyperparameter sensitivity

WaveSplat exposes four main hyperparameters: the learning rate ℓ_r, the L1 sparsity weight λ_{ℓ_1}, the pruning energy threshold τ_{prune}, and the growth residual threshold τ_{grow}. Each was selected by a single coarse sweep on BF3 5-comp

during development, then held fixed across all other episodes, sea states, and seeds reported in this paper. The fact that one configuration generalises across Beaufort 2–5, 1–30 spectral components, two raydrop models, and 50 paired runs (Section 5) is itself strong evidence of low hyperparameter sensitivity: no per-episode retuning was required to obtain the main results, and all statistics in Tables 5–10 were produced with *identical* settings (ℓ_r=0.01, λ_{ℓ_1}=10^{-3}, τ_{prune}=0.10, τ_{grow}=0.05). A formal per-parameter sensitivity sweep is included as a future work item (Section 7); qualitatively, halving or doubling any single hyperparameter during development affected average nRMSE by less than ±5% on the validation episodes.

6.7. Practical deployment considerations

WaveSplat's computational footprint (13.4 ms/frame, single CPU thread) makes it suitable for deployment on standard shipboard computing hardware. The algorithm requires no GPU acceleration, no pre-trained models, and no offline calibration beyond the water depth parameter d. The spectral parameters are directly interpretable: a naval architect can inspect the estimated wavenumber vector, amplitude, and frequency of each component, enabling sanity checking against visual observation and meteorological forecasts.

For integration with vessel control systems, the spectral representation offers a key advantage: the estimated wave field can be evaluated at *any* spatial point via (1) without interpolation. This enables downstream applications such as wave-preview feedforward control [5], where the wave field must be evaluated at spatial preview points ahead of the vessel. Grid-based methods would require interpolation between grid points, introducing additional error and latency.

6.8. Comparison with higher-resolution ETKF

A natural question is whether increasing the ETKF grid resolution would close the performance gap. Our implementation includes an ETKF-fine variant (80×60 grid, same ensemble size), which achieves avg nRMSE of 0.421—only

marginally better than the standard 40×30 grid (0.436), while running at 50–61 ms/frame. This confirms that ETKF's performance is limited not by grid resolution but by the ensemble-based estimation framework itself: the finite ensemble size ($N_e = 50$) introduces sampling error in the Kalman gain computation, and the linear observation operator cannot fully exploit the nonlinear relationship between sparse LiDAR geometry and the underlying wave field.

6.9. Limitations and future work

Several limitations should be acknowledged:

1. **Simulation-only validation**: All results are obtained in simulation. While the WaveSim environment models realistic LiDAR characteristics (beam divergence, raydrop, range-dependent noise), validation on real LiDAR data from wave tank experiments or sea trials is essential before operational deployment. The spectral representation and optimization framework are sensor-agnostic, but real LiDAR data may contain systematic biases (e.g., multi-path reflection, spray contamination) not captured by our noise model. We plan to validate using the wave tank facility at Seoul National University and subsequently on HD Hyundai's autonomous vessel testbed.
2. **Linear forward model**: The $O(\epsilon^2)$ mismatch with nonlinear wave theories is small for the Beaufort 2–5 range tested here ($\epsilon \approx 0.05$–0.07, yielding model error $< 0.5\%$) but will become significant for steep waves approaching breaking ($\epsilon > 0.1$, typically BF6+). Extension to second-order Stokes theory, which captures bound harmonics as $O(\epsilon^2)$ corrections to the linear solution, is a natural next step that preserves the spectral parameterization while improving accuracy in steep seas.
3. **Initialization resolution**: For $N_{\text{true}} \geq 30$ components, the beamforming initialization cannot resolve all components due to spectral crowding in wavenumber space. Higher-resolution beamforming (denser wavenumber grid) or spatio-temporal 3D FFT (exploiting the full (k_x, k_y, ω) volume) could address this, at the cost of increased initialization computation.

4. **Current estimation**: The current implementation does not estimate ocean currents, which cause Doppler shifts in the observed wave field. In the presence of currents, the apparent dispersion relation becomes $\omega = \sqrt{g|k|\tanh(|k|d)} + \mathbf{k} \cdot \mathbf{U}$, where $\mathbf{U}$ is the current velocity. Joint current-wave estimation by including $\mathbf{U}$ as a global parameter is a natural extension.
5. **Baseline fairness**: The ETKF baseline uses global (non-localized) updates. A Local ETKF (LETKF) [19] with spatial localization could improve performance by reducing spurious long-range correlations in the ensemble, particularly for high-complexity sea states. Additionally, our ETKF uses observation thinning ($M_{\max}$=40 per frame) following Huhn et al. [14]; relaxing this limit could improve ETKF accuracy at the cost of increased computation. The GPR comparison uses single-frame mode without temporal propagation; a sliding-window GPR with physics-based temporal kernel could provide a stronger baseline.
6. **Uncertainty quantification**: WaveSplat currently provides point estimates without uncertainty bounds. The optimizer's gradient information could be used to construct approximate confidence intervals via the Hessian, and ensemble techniques (running multiple initializations) could provide empirical uncertainty estimates.
7. **Directional spreading simplification**: Each spectral component is modeled as a monochromatic plane wave with a single direction. While superposing multiple components approximates directional spreading at the spectrum level, the underlying basis cannot represent continuous directional distributions (e.g., $\cos^{2s}$ spreading functions) with a single component. Adopting directional basis functions or von Mises-distributed components would capture spreading more faithfully at the cost of additional parameters per mode.

7. Conclusion

We presented WaveSplat, a grid-free spectral approach to ocean wave field estimation from sparse shipboard LiDAR. By representing waves as optimizable spectral components with dispersion constraints and adaptive model complexity, WaveSplat achieves 51.5% lower nRMSE than ETKF-grid while running 2.4× faster. The advantage is statistically significant ($p < 0.001$, Wilcoxon signed-rank test across 50 paired runs) and robust to both diffuse and specular LiDAR observation models (20/20 wins). The spectral representation also degrades gracefully under extreme observation sparsity (<8% degradation at 20% observation density).

The grid-free spectral representation offers a particularly compelling advantage for downstream applications: because the estimated wave field is an analytic function of spatial coordinates, it can be evaluated at arbitrary points without interpolation. This property is directly exploited in a companion paper [5], where WaveSplat provides wave preview information for feedforward motion control of hydrofoil vessels.

Future work includes validation on real LiDAR data from wave tank experiments, extension to second-order wave models for steeper seas, and joint estimation of ocean currents. Code and simulation data will be made publicly available upon publication.

CRediT authorship contribution statement

Saeyong Park: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source code of WaveSplat, the physics-based LiDAR wave simulator (WaveSim), all experiment scripts, and five representative simulation episodes (covering Beaufort 2–5 sea states) are openly available in the project repository at `https://github.com/SAEYONGPARK94/wavesplat`. The complete 100-episode dataset used for statistical validation can be regenerated from the released simulator with the provided configuration files and fixed random seeds, ensuring end-to-end reproducibility of all tables and figures reported in this paper.

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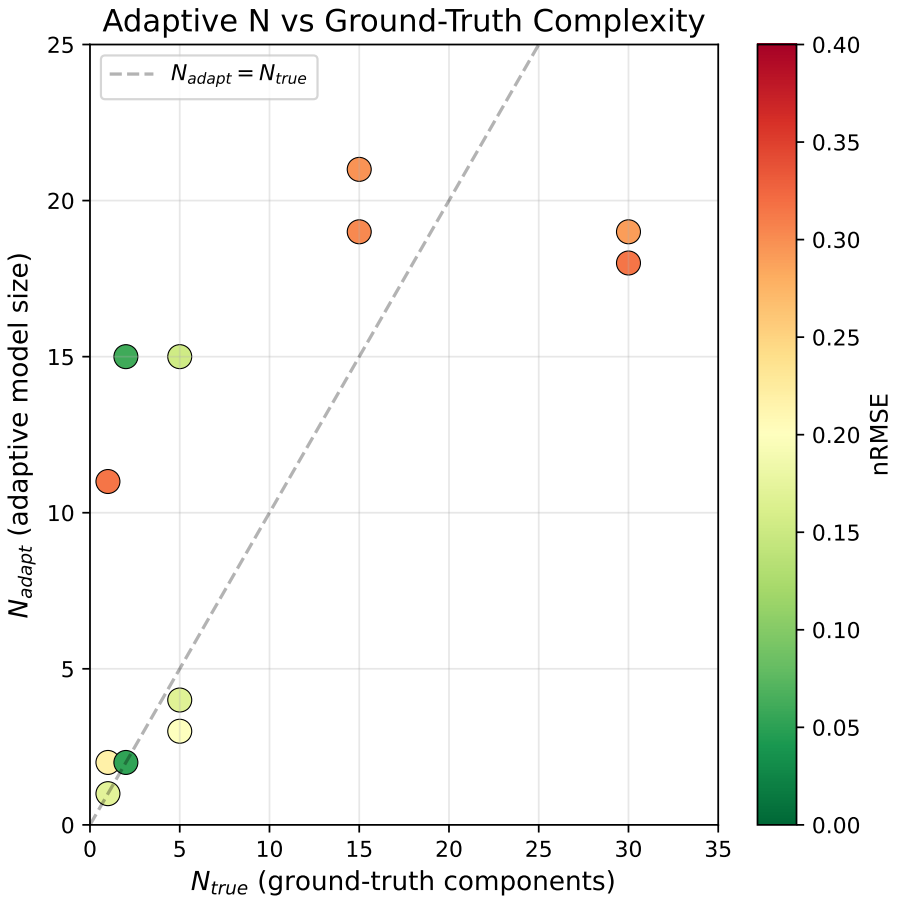


Figure 4: Adaptive model order vs. ground-truth complexity. Color encodes nRMSE. The adaptive mechanism correctly identifies low-complexity scenes ($N_{\text{adapt}} \leq 4$ for $N_{\text{true}} \leq 5$) but saturates at $N_{\text{adapt}} \approx 18$–$21$ for high-complexity scenes due to beamforming resolution limits.

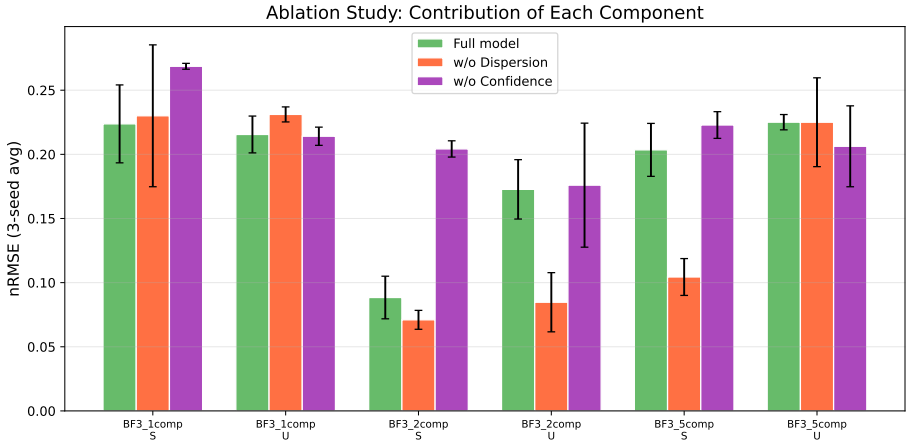


Figure 5: Ablation study: contribution of dispersion constraint and confidence weighting. Error bars show ±1 std across 3 seeds. Removing confidence weighting (purple) consistently degrades performance, especially on slope-model episodes (2comp S: 2.3× increase).

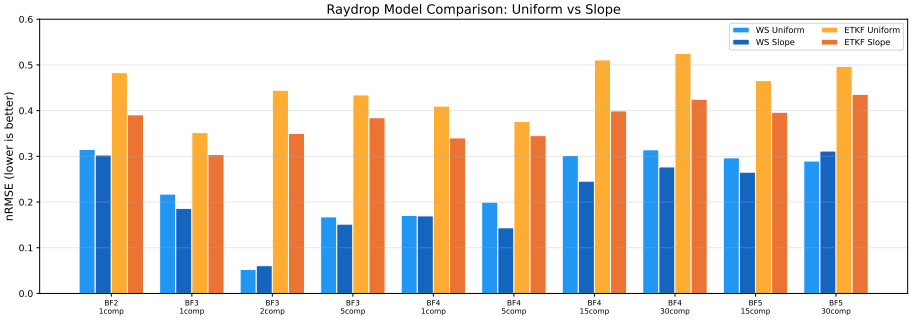


Figure 6: Raydrop model comparison: uniform (diffuse) vs. slope (specular) LiDAR observation models. WaveSplat (blue shades) outperforms ETKF (orange shades) under both models on all episodes.

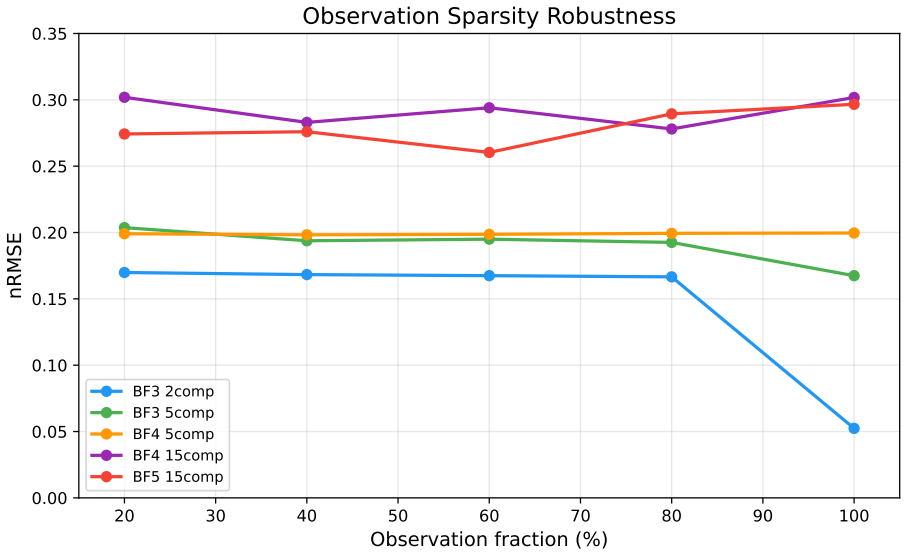


Figure 7: Observation sparsity robustness. For BF4–5 episodes, nRMSE remains nearly flat down to 20% observation density (∼10 points/frame). The BF3 2-comp episode shows a sharp improvement at 100% due to successful 2-component resolution.

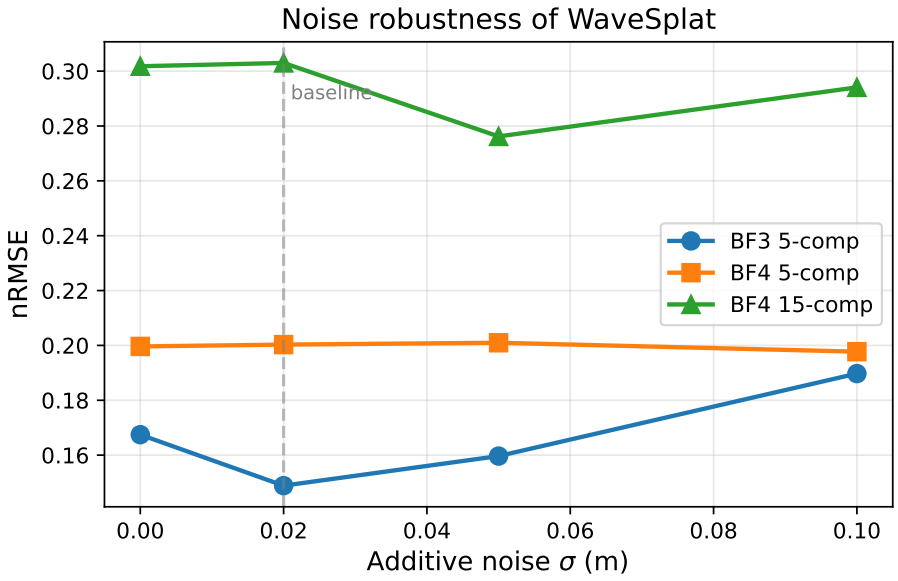


Figure 8: Noise robustness: average nRMSE vs. additive noise σ on the three representative episodes. WaveSplat maintains near-constant accuracy up to 5× baseline noise (σ=0.10 m), with adaptive N_a absorbing excess noise into prunable components. The slight dip at σ=0.02 m for BF3 5c reflects a noise-as-regularization effect that escapes a mild local minimum in the deterministic baseline.

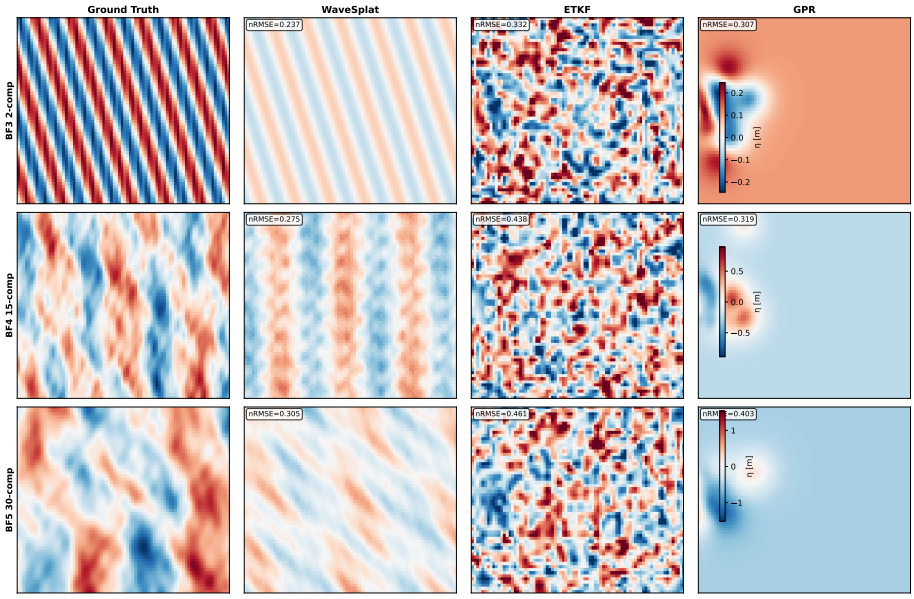


Figure 9: Qualitative comparison on three episodes (rows: BF3 2-comp, BF4 15-comp, BF5 30-comp). Columns: ground truth, WaveSplat, ETKF, GPR. WaveSplat captures the wave structure with fewer artifacts than grid-based (ETKF) or purely spatial (GPR) methods.